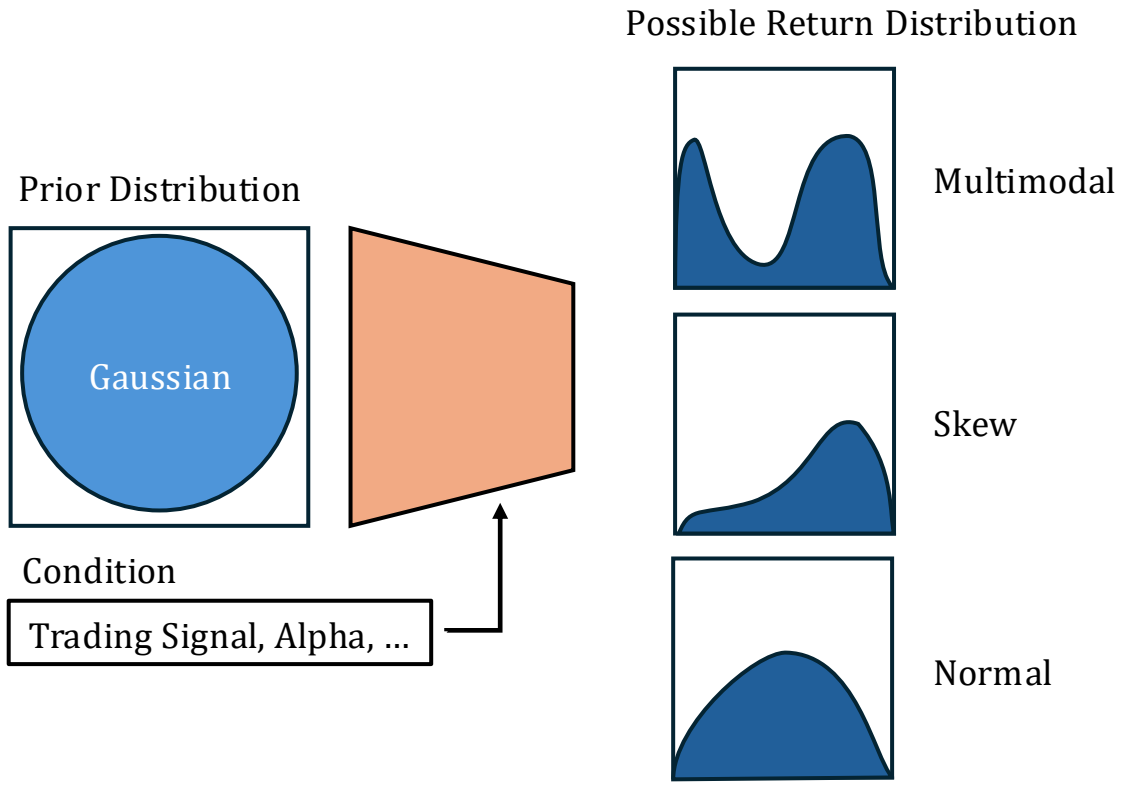
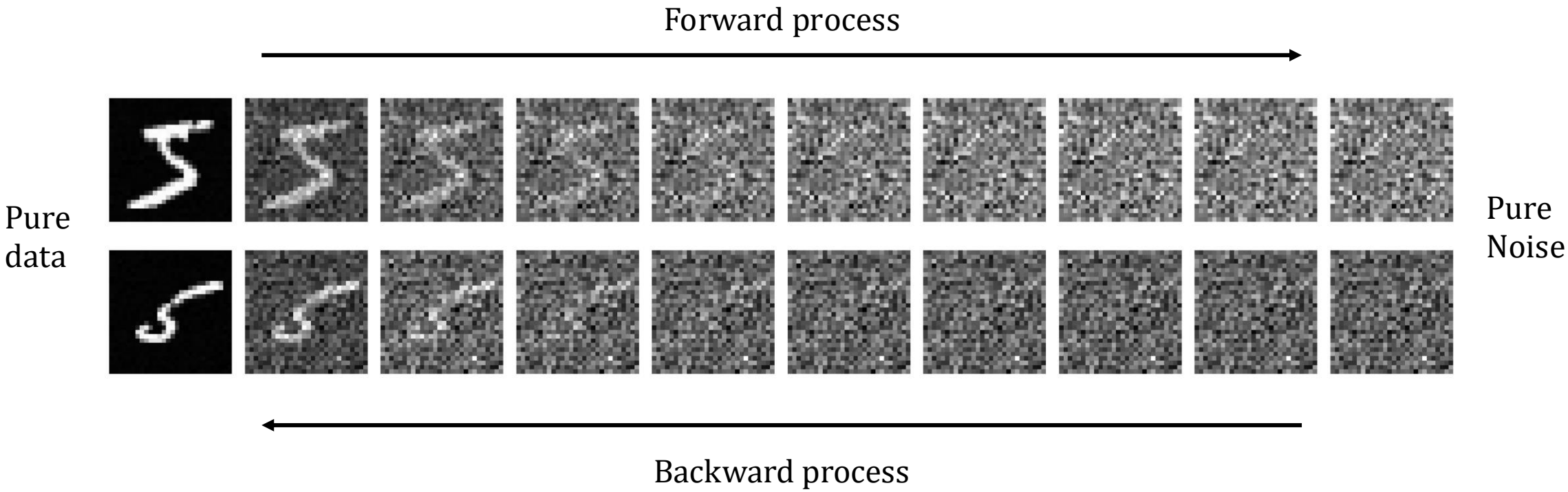
Generative Model

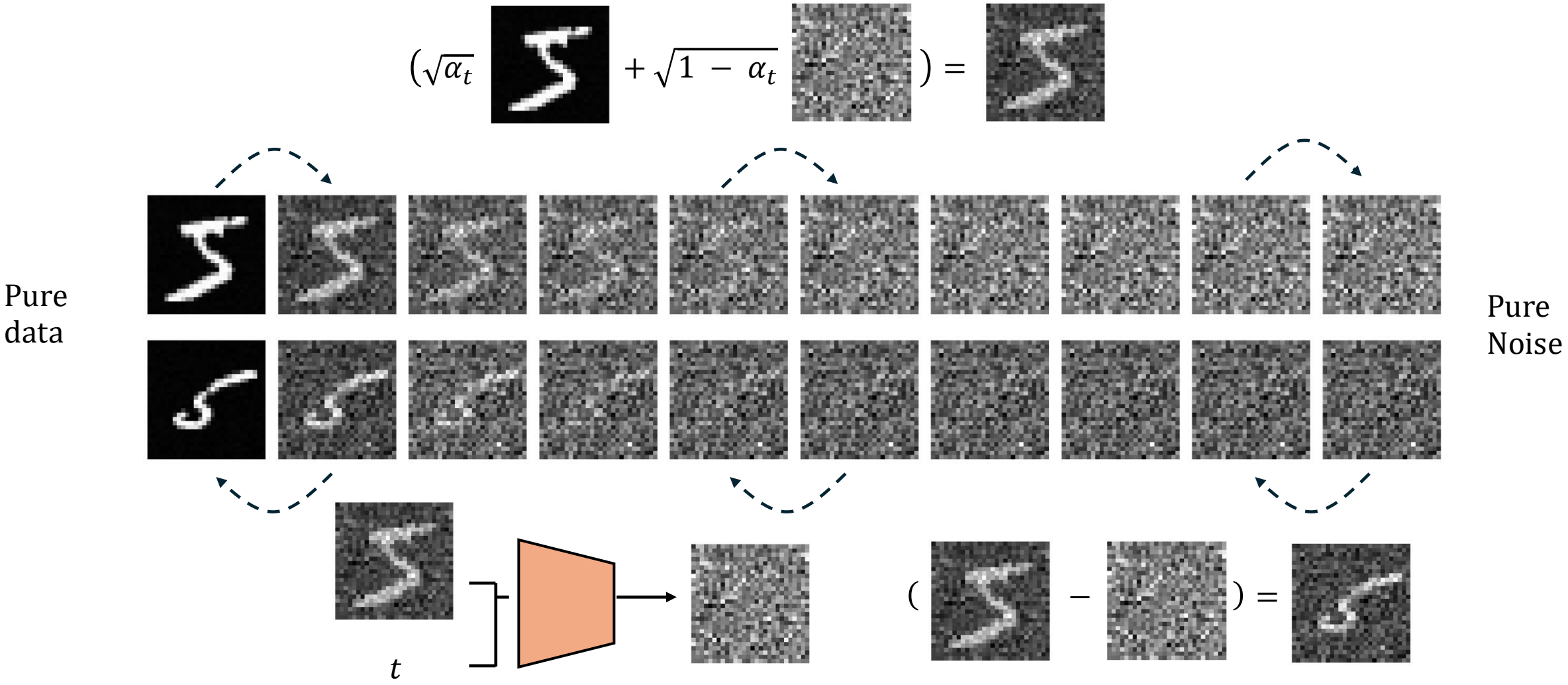
Give a signal, we can actually **“See” the return probability.**



The theory behind diffusion

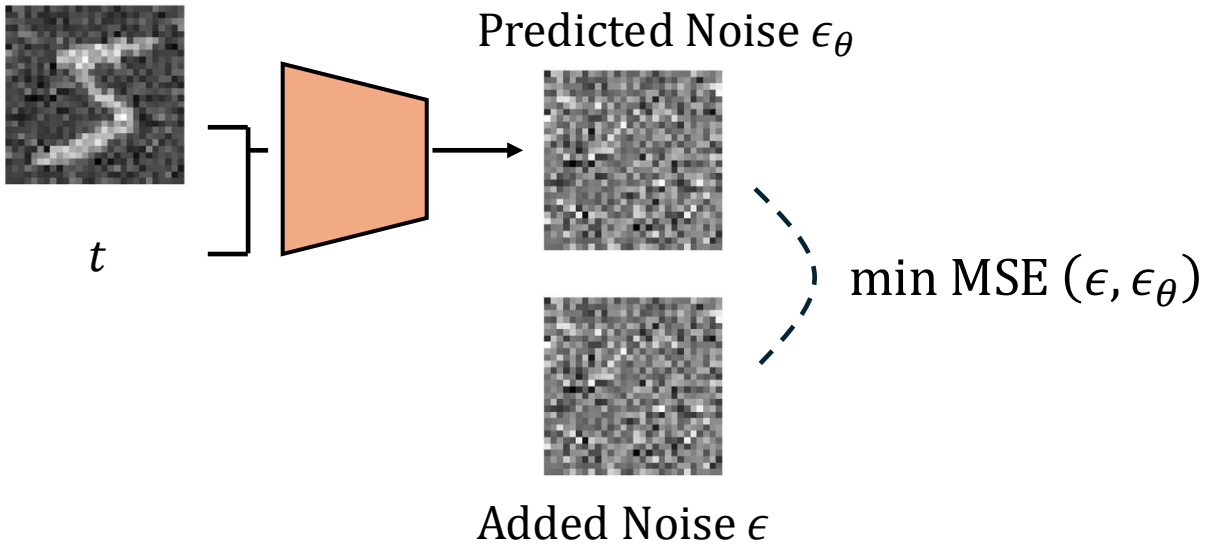


The theory behind diffusion



The theory behind diffusion

Learning Objective

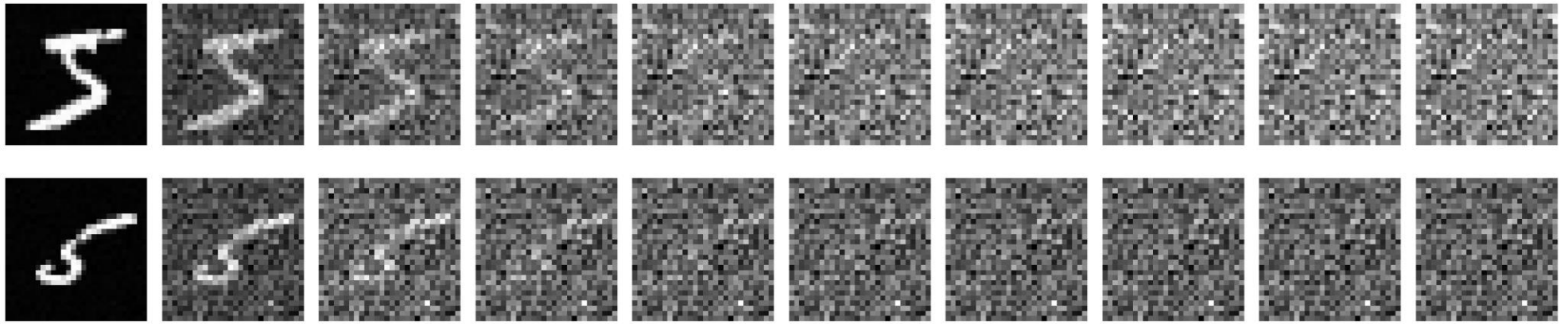


Model doesn't directly learn the label itself but the added noised. The added noise is actually a score function for the target distribution.

The theory behind diffusion

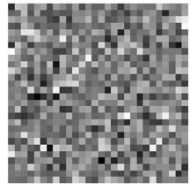
Forward process : $dx = f(x,t)\,dt + g(t)dw$

Pure data



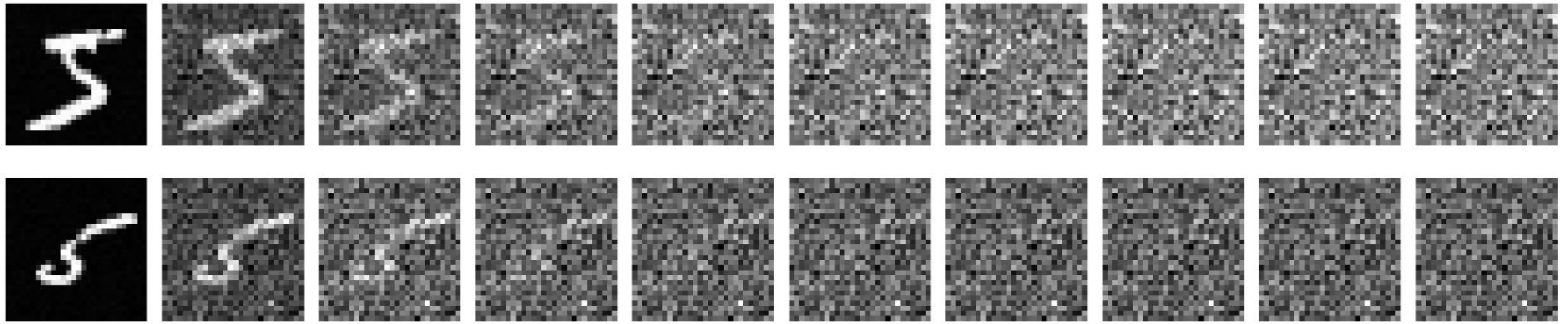
Pure Noise

Backward process : $dx = \left[f(x,t) \;\; - g(t)^2\,\nabla_{x_t} \log p(x_t)\right] d\tilde{t} + g(t)\,d\tilde{w}$



The theory behind diffusion

Forward process : $dx = f(x,t)\,dt + g(t)dw$



Pure data

Pure Noise

Backward process : $dx = \left[f(x,t) \;\; - g(t)^2\,\nabla_{x_t} \log p(x_t)\right] d\tilde{t} + g(t)\,d\tilde{w}$

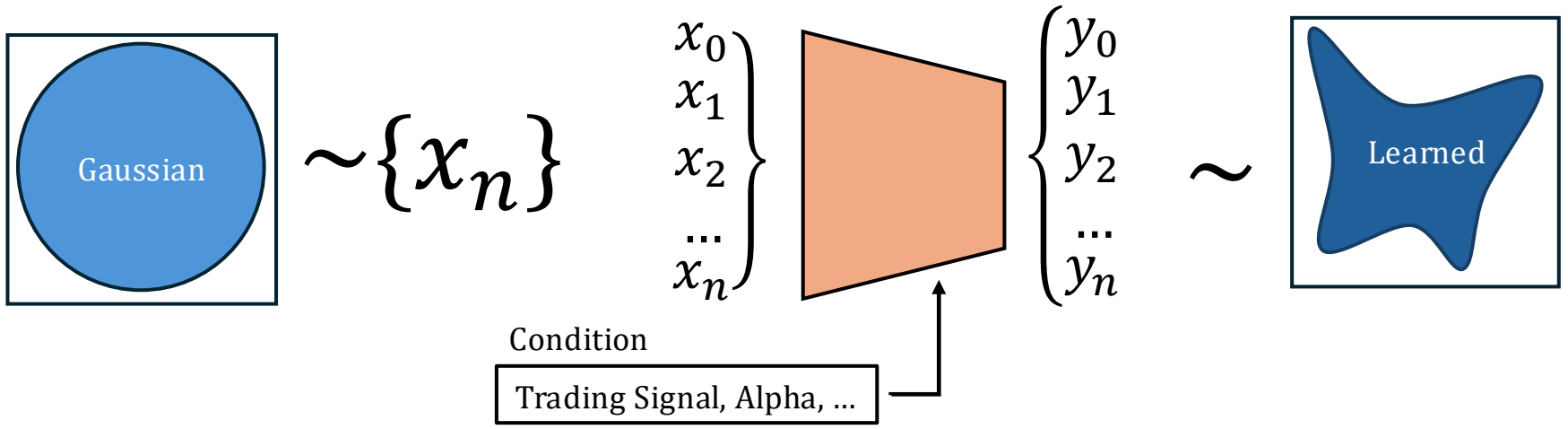
This is called score-function, thus the method is called score-based generative model

And we are fitting the system distribution with score function.

Inference

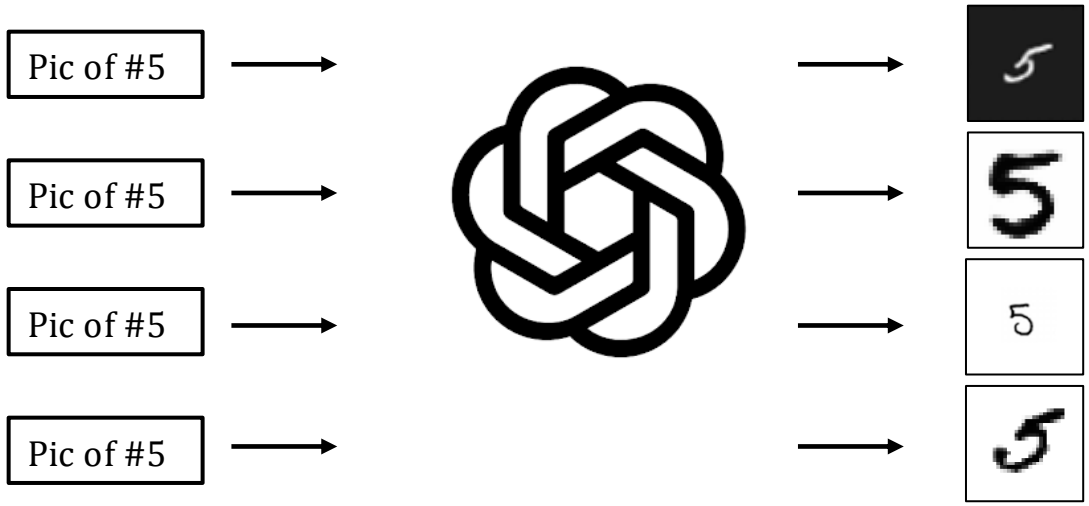
How do we get the ”distribution”?

We sample multiple times from the prior using the same condition. The collection of outputs forms our predicted target distribution.



Inference

This is conceptually similar to prompting GPT to generate images with the number "5" a thousand times. The resulting images form a distribution of the concept "5".

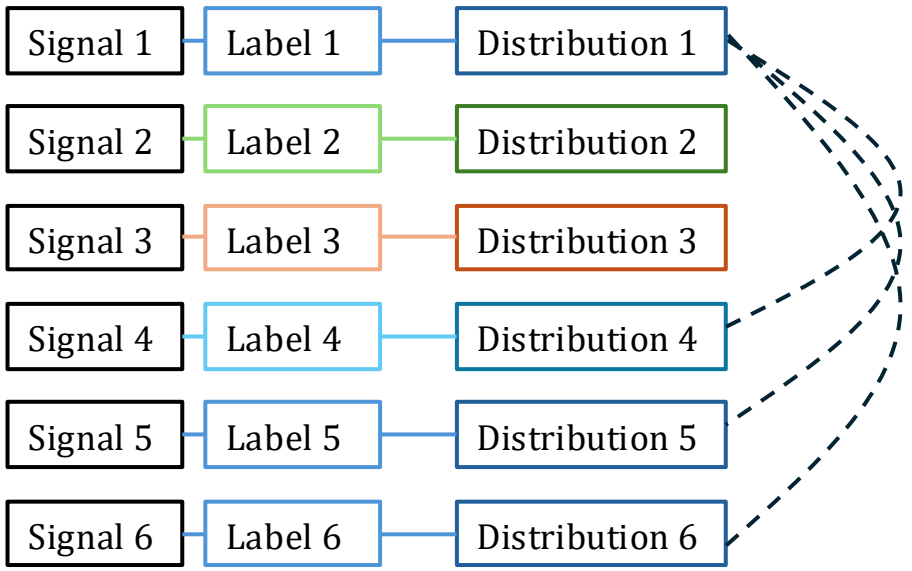




distribution of the concept "5"

Metrics

In practice, we don't observe the true distribution. Instead, for each signal, we observe its label y_n and predict a distribution P_n.

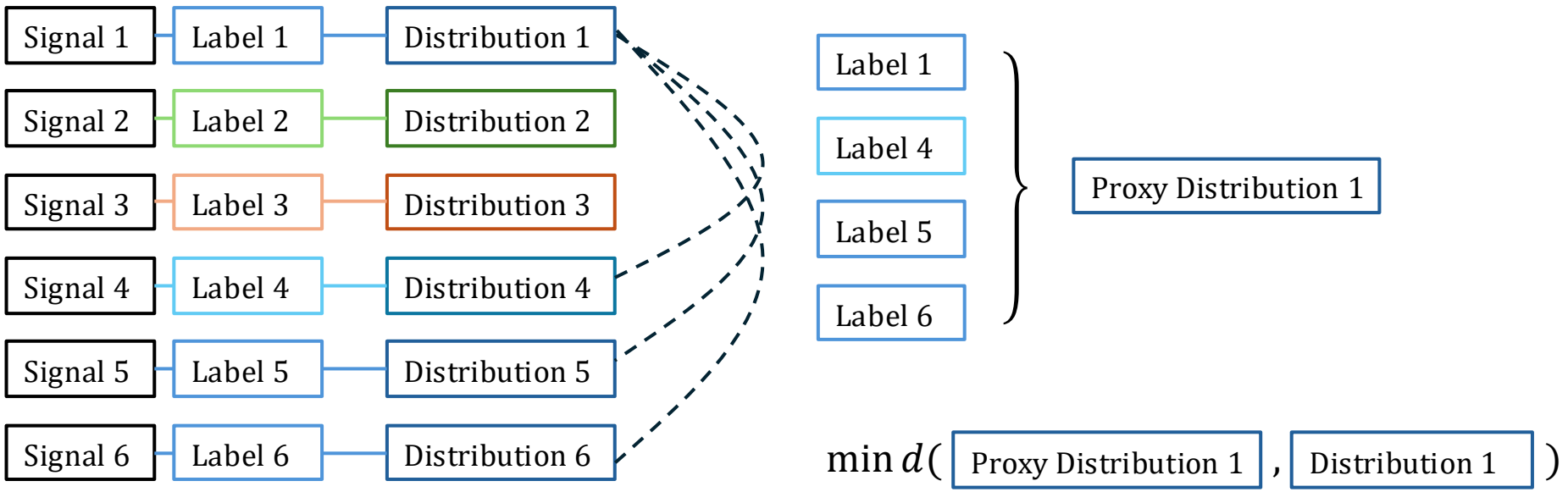


The distance between these distributions is close.

This means that if we trade distribution 1, then we also trade distribution 4, 5, 6. Because they have the similar distribution.

Metrics

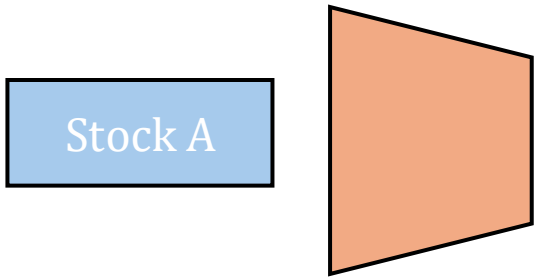
In practice, we don't observe the true distribution. Instead, for each signal, we predict a distribution P_n, and observe its label y_n



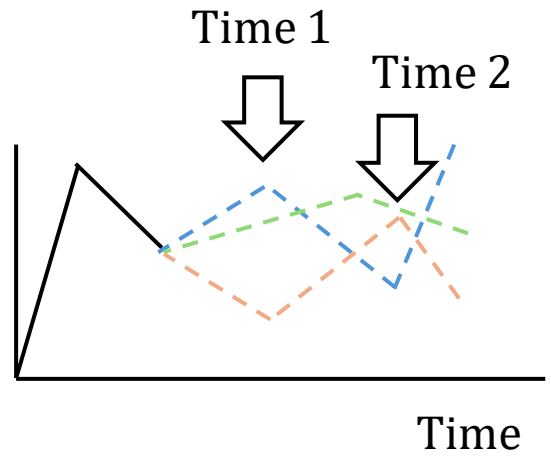
Therefore, we can form a **proxy distribution** by collecting all the similar distribution's observation and compare the proxy distribution with the predict distributions.

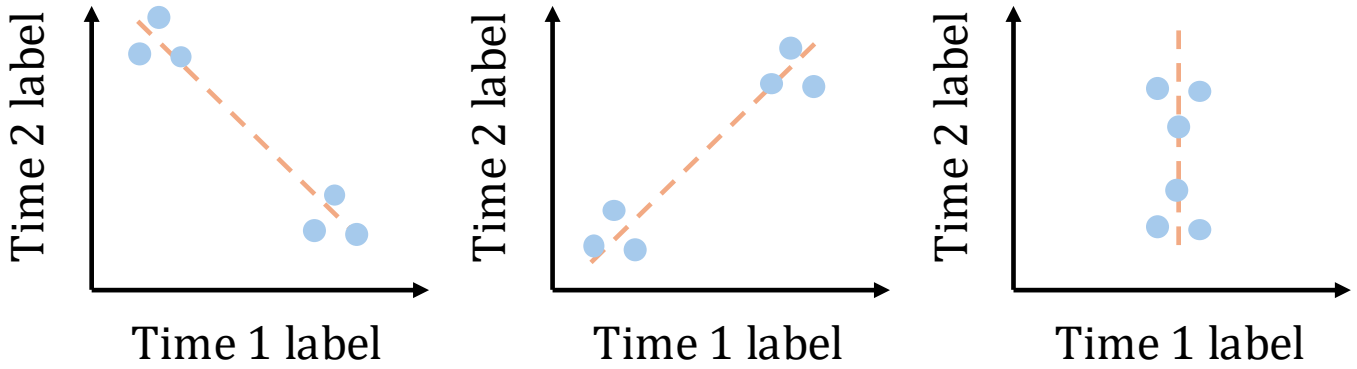
Ongoing MM Research

Different label time zone



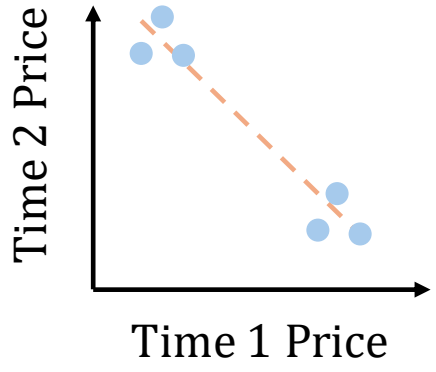
$[label_1, label_2, \ldots, label_n]$

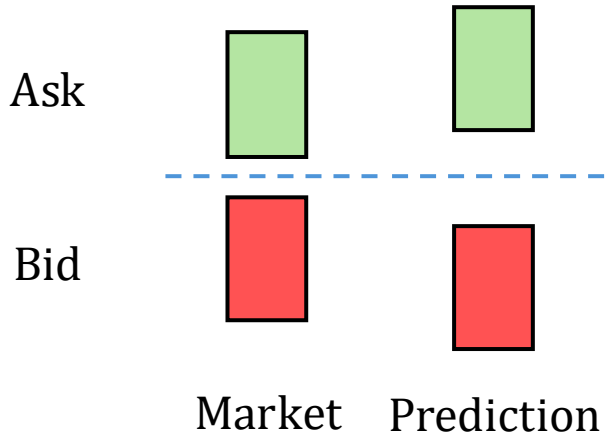




Base on the prediction, we can perform multiple strategy

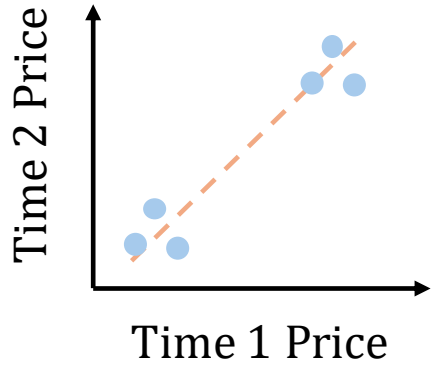
Ongoing MM Research

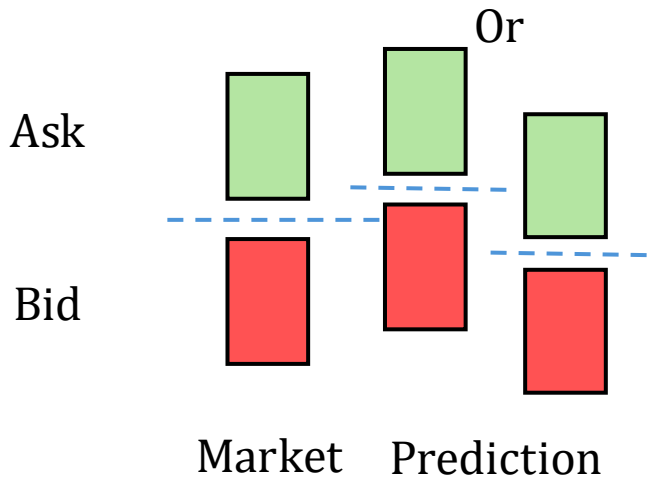




The short-horizon prediction moves against the long-horizon trend.

Widen spreads and rely on inventory reversion.

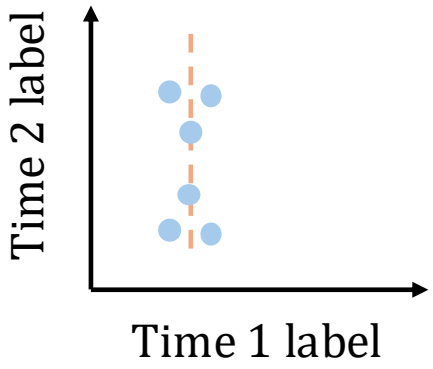


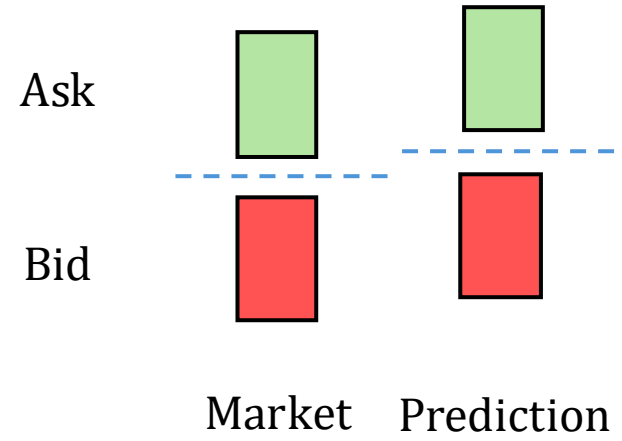


The short-horizon prediction is consistent with the long-horizon trend.

Shift from pure market making toward directional skewing

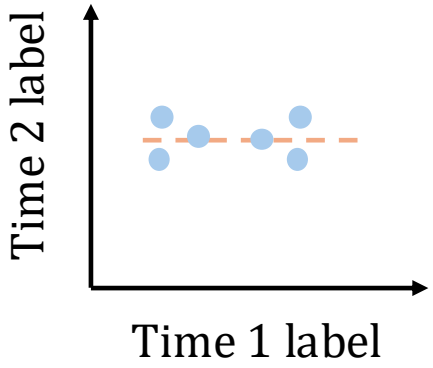
Ongoing MM Research

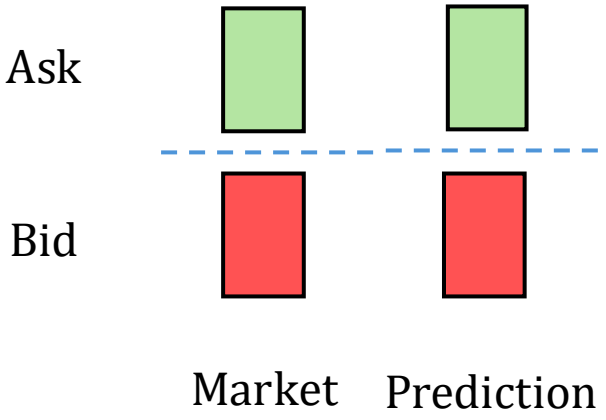




The short-horizon prediction indicates an imminent short-term price movement.

Take a directional position aligned with the short-term signal.





The long-horizon prediction reflects slower dynamics and structural uncertainty.

We keep the spread and minimizing the directional exposure.

Cross-Equity Research

Co-movement Research

